

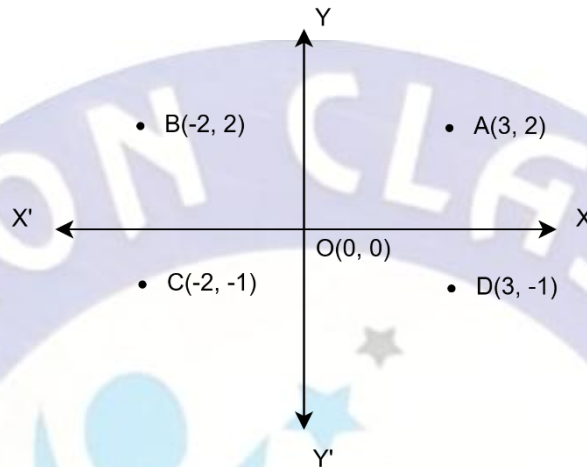
MOTION

- **Position:**

- The coordinate of an object is called its position.

OR

- The position of an object/body/particle which denotes that the object is in either rest or motion.
- It can be positive, negative or zero.



- It is one dimensional, denoted by X.

Here,

The position of A = (3, 2)
The position of B = (-2, 2)
The position of C = (-2, -1)
The position of D = (3, -1)

- **Motion**

- The position of an object changes with time is called the object in motion.

For example:-

- ✓ The moving bikes, cycle, bus, the walking man, the running man etc.

- **Rest**

- When the position of an object does not change with time, it is called the object is in rest.

For example:-

- ✓ A sitting person.

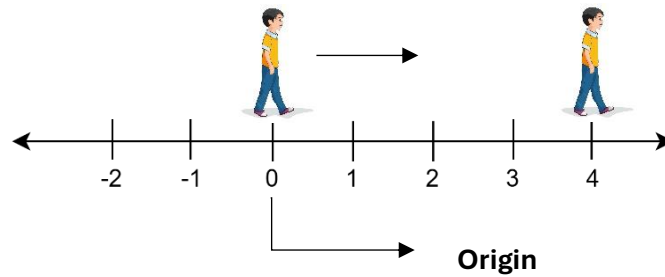
Note:

- Motion $\neq$ Rest

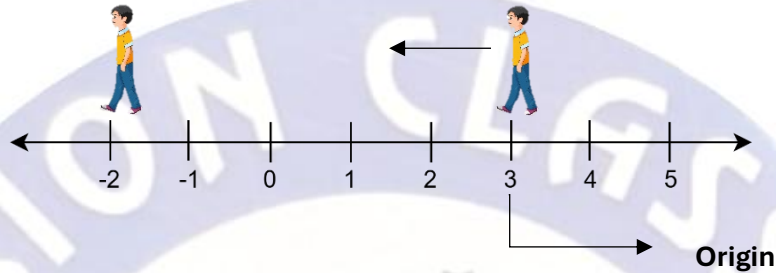
- **Reference Point / Origin**

- When an object moves from that point is called the origin of that object or reference point.

▪ **Case 1:**



▪ **Case 2:**



• **Distance (D)**

- The length of path travelled by an object is called distance of that object.
- Distance is a scalar quantity, i.e., it has only magnitude.
- It is always positive i.e., It can't be negative.
- It can't be zero (0).

• **Displacement (S)**

- It is the shortest distance (length) between initial and final points.
i.e,

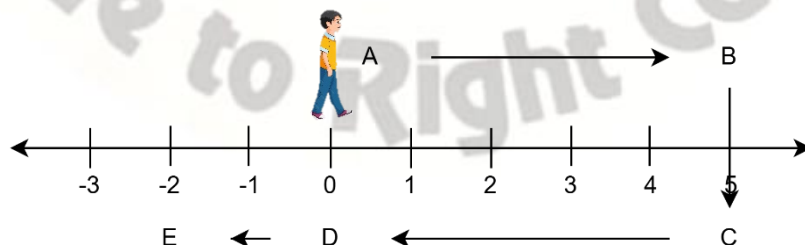
$$\text{Displacement (s)} = \text{Final position} - \text{Initial position}$$

- It is a vector quantity, i.e., it has magnitude as well as direction.
- It can be negative, positive, or zero.

For example:-

Q. Find distance and displacement (S).

Ans



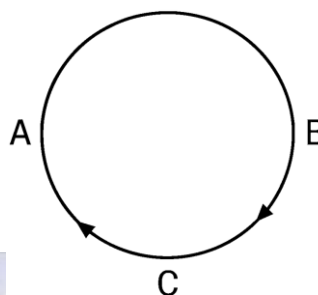
$$\Rightarrow \text{Distance} = 5 + 5 + 2 \text{ [AB + CD + DE]} \\ = 12 \text{ meters}$$

$$\Rightarrow \text{Displacement} = -2 - 0 = -2 \text{ meters}$$

Q2. A person goes in a circular park of radius 28cm with ACB. Find (i) distance, (ii) displacement.

Ans

(i) Distance $= \pi \cdot r$
 $= \frac{22}{7} \times 28$
 $= 88 \text{ cm}$



(ii) Displacement $= \overline{AB} = \text{diameter}$
 $= 2 \times r$
 $= 2 \times 28 = 56 \text{ cm}$

DIFFERENCE BETWEEN DISTANCE & DISPLACEMENT

Distance (d)	Displacement (s)
It is actual path of length.	It is shortest distance between two points.
It is a scalar quantity.	It is a vector quantity.
It is not directional.	It is directional.
It can't be negative or zero.	It can be negative, positive, or zero.
It is always positive.	It is positive, negative, or zero.
It is denoted by 'D'.	It is denoted by 'S'.

Special Note:

1. In a straight path, the distance and displacement both have equal in magnitude.
2. Rest and motion of a body only exist with reference point, otherwise It can't be defined.

TYPES OF MOTION OF A BODY

▪ Case 1: On the basis of path

- i) Linear motion.
- ii) Circular motion.
- iii) Oscillatory motion.
- iv) Random motion.
- v) Rectilinear motion.
- vi) Vibrating motion.
- vii) Instantaneous motion.
- viii) Rotary motion.
- ix) Translatory motion.

▪ Case 2: On the basis of duration

- i) Periodic motion.
- ii) Non – Periodic motion.

▪ **Case 3: On the basis of Speed**

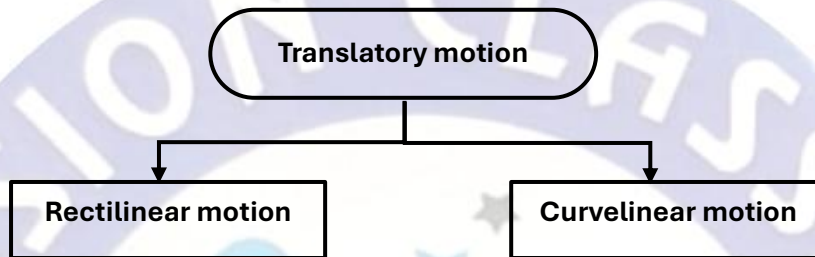
- i) Uniform motion.
- ii) Non - Uniform motion.

• **Linear Motion**

- When an object travels either in a straight path or a curve path, then the motion is called linear motion.

For example:-

- ✓ The motion of bullet, the motion of sliding body, the motion of bike, car, bus, train, aeroplane etc.



1. **Rectilinear motion**

- When an object travels in a straight path, then the motion is called rectilinear motion.

For example:-

- ✓ The motion of a bullet, The motion of an elevator, the free falling body.



STRAIGHT PATH

2. **Curvilinear motion**

- When an object travels in a curved path, then the motion is called curvilinear motion.

For example:-

- ✓ When motorcars travel on curved roadways.
- ✓ A person who plays ice-skating.
- ✓ The projected ball.



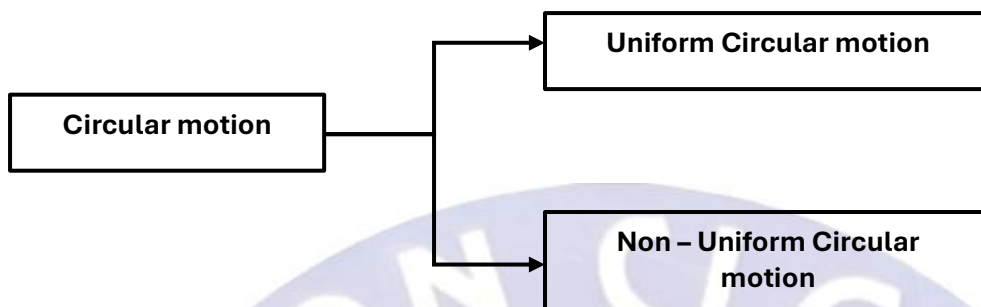
CURVED PATH

• **Circular motion**

- When an object moves in a circular path, then the motion is called circular motion.

For example:-

- ✓ The motion of the moon around the earth, the motion of the earth around the sun, the motion of the needle of a watch, the motion of satellite around the earth, a stone tied to a string, the motion of wheels of vehicles, etc.



- **Oscillatory motion**

- The to and fro motion of an object from its mean position is called oscillatory motion.
- It is a type of periodic motion.

For example:-

- ✓ The motion of pendulum of watch, swinging of swing, vibrating strings in a guitar, freely hanging bob, etc.

- **Random Motion**

- The motion of an object with no specific path is called random motion.

For example:-

- ✓ The motion of flying bee, butterfly, fish swimming in water, motion of birds, motion of particles of perfume, etc.

- **Vibrating Motion**

- It is a type of oscillatory motion.

For example:-

- ✓ Tuning fork, vibration of strings of a guitar, vibration of the vocal chord in our throat while producing sound, vibration of ear drum.

- **Instantaneous Motion**

- The motion of an object at a particular moment in a time is called instantaneous motion.

For example:-

- ✓ The speed of a bike, car, bus etc shows by speedometer at a time.

- **Rotary Motion**

- That motion in which an object moves about a fixed axis.

For example:-

- ✓ The motion of the wheel, gears, motors throwing which cricket ball, etc.

- **Translatory Motion**

- The motion in which all points of a moving body move uniformly in the same line or direction is called translatory motion.
- Translatory motion is also called translational motion.

For example:-

- ✓ Pulling a block, car, door, etc.

- **Periodic Motion**

- The motion of an object repeated in equal time intervals is called periodic motion.

For example:-

- ✓ Motion of the needle of a sewing machine, the motion of swing, etc.

Note:

- Non Periodic motion $\neq$ Periodic motion

- **Speed**

- The rate of change of distance with respect to time is called speed.

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

- It is a scalar quantity.

- The S.I unit of speed = $\frac{\text{the S.I unit of distance}}{\text{the S.I unit of time}} = \text{m/s.}$

∴ The S.I unit of speed is m/s.

Note:

1. Speed $\propto$ Distance
2. Speed $\propto \frac{1}{\text{time}}$

Note:

1. To change m/s to km/h, we multiply by $\frac{18}{5}$.
2. To change km/h to m/s, we multiply by $\frac{5}{18}$.
3. Speed can't be negative.

- **Average Speed**

- The ratio of total distance travelled and total time taken is called average speed.

$$\therefore \text{Average Speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

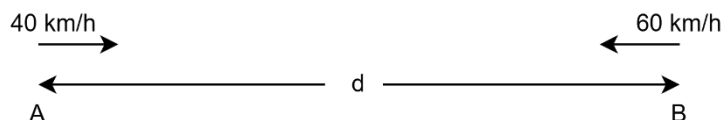
- Average speed is denoted by V_{avg} .

- The unit of average speed is denoted by km/h or m/s.

For example:-

- Q. If a bus travels 40 km/h from A to B and returns back from B to A with a speed of 60 km/h. Find the average speed in the whole journey.

Ans



Let,

Distance between A to B = d km

∴ Time taken to travel from A to B = $\frac{d}{40}$ hours

&

∴ Time taken to travel from B to A = $\frac{d}{60}$ hours

∴ Total time = $\frac{d}{40} + \frac{d}{60} = \frac{3d + 2d}{120} = \frac{5d}{120}$

∴ Average speed = $\frac{\text{Total distance}}{\text{Total Time}}$

$$\Rightarrow V_{\text{avg}} = \frac{d + d}{\frac{5d}{120}} = \frac{2d \times 120}{5d}$$

$$\Rightarrow V_{\text{avg}} = 2 \times 24 = 48 \text{ km/h}$$

• Uniform Speed

- If a body travels equal distance in equal time intervals then we said that the body is in uniform speed.
- If a body travels s distance in t time intervals.

$$\therefore \text{Speed} = \frac{\text{Displacement}}{\text{Time}}$$

$$\Rightarrow V = d/t \quad \Rightarrow d = v \cdot t$$

For example:-

- ✓ The speed of fan's blade, the motion of earth – water motor pump etc.

• Non – Uniform speed

- If a body travels unequal distances in equal time intervals then the motion is called non – uniform speed.

For example:-

- ✓ The speed of car, bus, truck etc in non – uniform form.

• Velocity

- The rate of change of displacement with respect to time is called velocity.

$$\therefore \text{Velocity} = \frac{\text{Change in displacement}}{\text{time}} = \frac{\text{displacement}}{\text{time}}$$

$$\therefore \text{Velocity} = \frac{\text{final displacement} - \text{initial displacement}}{\text{time}}$$

Actually,

The velocity of an object is its displacement – per unit time.

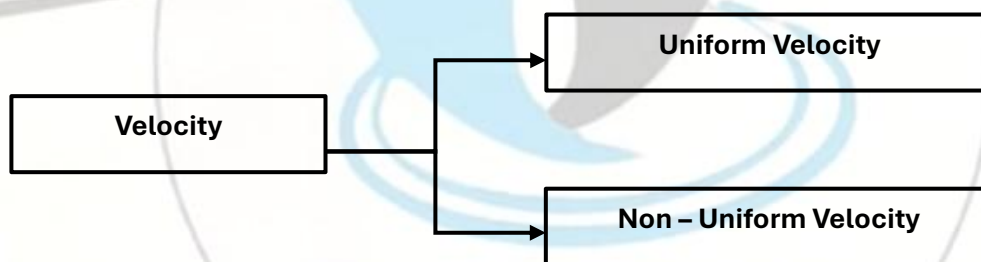
- ↳ It is a vector quantity.
- ↳ The S.I unit of velocity is m/s or km/h (greater).

- It may be positive, negative or zero.
- It is denoted by 'V'.

$$\text{Velocity} \propto \begin{matrix} \text{(i) displacement} \\ \text{(ii) } \frac{1}{\text{time}} \end{matrix}$$

DIFFERENCE BETWEEN SPEED AND VELOCITY

Speed	Velocity
Speed = distance / time	Velocity = displacement / time
It is a scalar quantity	It is a vector quantity
It can't be negative or zero	It can be negative, positive, or zero
The average speed of an object can't be zero	The average velocity of an object can be zero
It is rate of change of distance	It is rate of change of displacement



1. Uniform Velocity

- If the velocity of an object does not change with time, it is called uniform velocity.
OR
- If the velocity of an object travels equal displacements in equal intervals of time, it is called uniform velocity.
- The speed and direction of an object is constant i.e, object is moving in a straight line with fixed speed.

Note:

- In Uniform Velocity the acceleration of an object is zero.

• Non - Uniform Velocity

- If the velocity of an object travels unequal displacements in equal time interval, then the motion is called non- Uniform velocity.

• Acceleration

- The rate of change velocity with respect to time is called acceleration.

$$\therefore \text{Acceleration} = \frac{\text{Change in Velocity}}{\text{Time}}$$

$$\Rightarrow \text{Acceleration} = \frac{\text{Final Velocity} - \text{Initial Velocity}}{\text{Time}}$$

If final velocity = v
Initial velocity = u
& time taken = t

- Acceleration is denoted by $\vec{a}$ / a .

$$\therefore a = \frac{v - u}{t}$$

- It is a vector quantity.
➤ The S.I unit of acceleration is m/s^2 .
i.e,

$$\frac{\text{The S.I unit of Velocity}}{\text{The S.I unit of Time}} = \frac{\text{m/s}}{\text{s}}$$

$\therefore$ The S.I unit of $\vec{a}$ is m/s^2 .

Special Note:

- Velocity $\propto$ Acceleration
i.e, $V(\uparrow)$ then $a(\uparrow)$
 $V(\downarrow)$ then $a(\downarrow)$
- For Uniform Velocity, $\vec{a} = 0$

For Example:-

- i) When an object comes to motion from rest.
ii) When a ball is thrown.

• **Retardation**

- When velocity of an object decreases then acceleration of the object also decreases, the decreasing acceleration is called retardation.
i.e,

$$\vec{a} / a = -ve$$

- Retardation is equal in magnitude to acceleration but opposite in direction.

For Example:-

- ✓ When we apply brakes in vehicle, when a stone thrown upwards.

Note:

- The negative acceleration is called retardation.

- ✓ Retardation is also called deceleration.
- ✓ When velocity increases then acceleration also increases, this increasing acceleration is called acceleration.

i.e,

acceleration = +ve

- Average acceleration

- The ratio of change in velocity and change in time is called average acceleration.

$$\therefore \text{Average acceleration} = \frac{\text{Change in Velocity}}{\text{Change in Time}}$$

- If the final velocity of an object = V_2
- If the initial velocity of an object = V_1
- If the final time of an object = t_2
- If the initial time of an object = t_1

Then,

$$\therefore \text{average} = \frac{V_2 - V_1}{t_2 - t_1}$$

- The S.I unit of average acceleration is also m/s^2 .

Note:

1. If velocity and acceleration having same direction then velocity increases.

i.e,

$$, \vec{v}(\rightarrow), \vec{a}(\rightarrow) \quad \text{then } V(\uparrow)$$

2. If velocity and acceleration having opposite direction then velocity decreases.

i.e,

$$\vec{V}(\rightarrow), \vec{a}(\rightarrow) \quad \text{then } V(\downarrow)$$

- Uniform acceleration

- That acceleration in which equal changes in velocity in equal time intervals, called uniform acceleration.

If time interval is t .

If initial velocity = u .

& final velocity = v .

$$\therefore \text{Acceleration} = \frac{\text{Change in Velocity}}{\text{Time interval}}$$

$$\therefore a = \frac{v - u}{t}$$

- **Non – Uniform acceleration OR Variable acceleration**

- That acceleration in which unequal changes in velocity in equal interval of time.

For Example:-

- ✓ A free falling ball, A bike/cycle moves on slider road etc.

- **Graph**

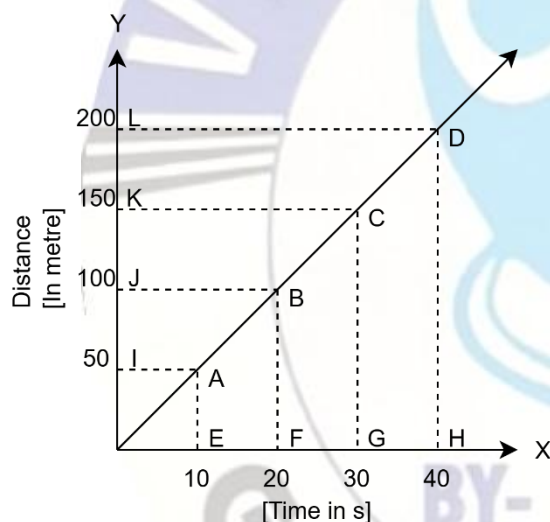
- A graph is a powerful method to represent the relation between two variable quantities.

Note:

1. Only base/fundamental quantities are represented on X – axis.
2. Only derived quantities are represented on Y – axis.

- **Case 1:**

- The distance – time graph of an object moving with a uniform speed is a straight line.



➤ Distance Covered by the Car at difference instant of time

Time (t)	Distance (m)
10	50
20	100
30	150
40	200

⇒ Speed = Slope of the graph

⇒ Speed = $\frac{\text{the distance travelled}}{\text{Time Taken}}$

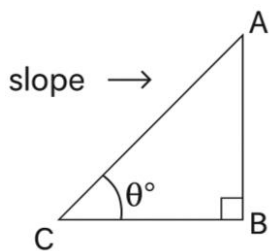
$$\therefore \text{Slope} = \text{speed} = \frac{AE}{OE} = \frac{BF}{OF} = \frac{CG}{OG}$$

$$= \frac{50}{10} = \frac{100}{20} = \frac{150}{30}$$

∴ Slope = speed = 5 m/s.

Note:

- **Slope / Inclination / $\tan \theta$**



$$\therefore \text{Slope} = \frac{\text{खड़ा}}{\text{पड़ा}} = \frac{y}{x}$$

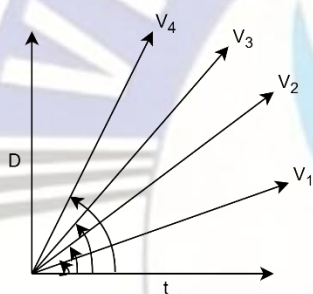
$$\Rightarrow \tan \theta = \frac{y}{x} = \frac{AB}{BC}$$

i.e.,

$$\text{Slope} \propto \text{खड़ा (y)} \propto \text{Speed}$$

$$\& \text{Slope} \propto \frac{1}{\text{पड़ा (x)}}$$

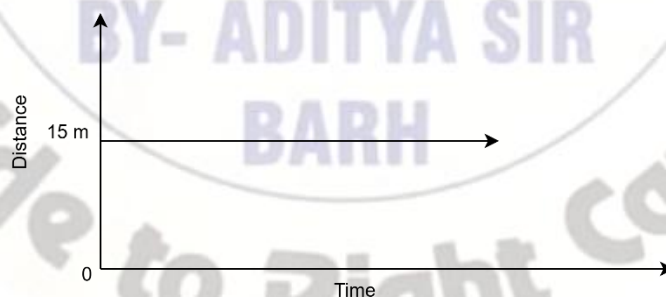
- The slope of distance – time graph gives speed.
- $\tan \theta$ is a trigonometrical ratios which more studies in higher classes.



$$\Rightarrow \text{Speed} = V_4 > V_3 > V_2 > V_1$$

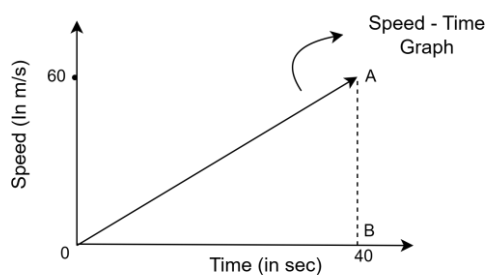
▪ **Case 2:**

- If the object is in rest then the distance – time graph is parallel to X – axis.



▪ **Case 3:**

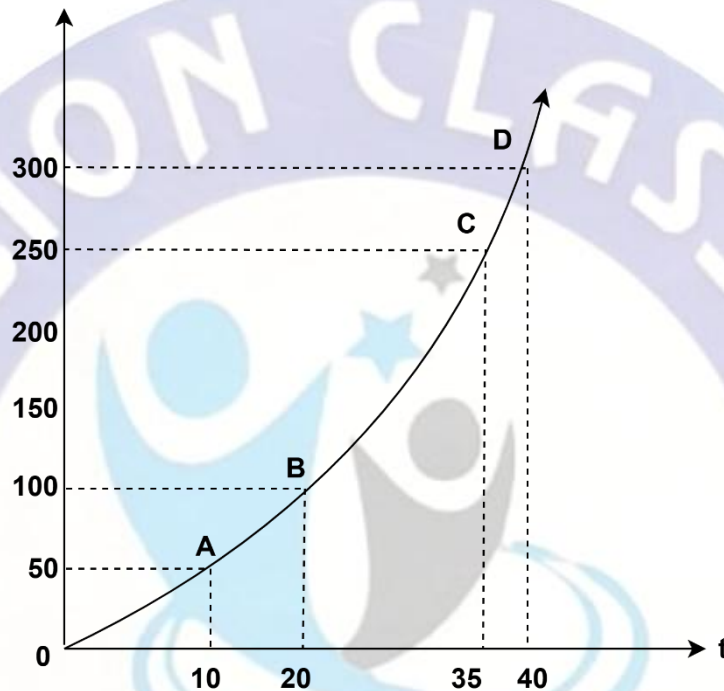
- The area of speed – time graph gives distance.



$$\begin{aligned}\Rightarrow \text{Distance} &= \text{Area of } \triangle AOB \\ &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times OB \times AB \\ &= \frac{1}{2} \times 40 \times 60 \\ \text{Distance} &= 1200 \text{ meter}\end{aligned}$$

▪ Case 4:

➤ For non – uniform speed the distance – time graph is a curve line.



Here,

$$\text{Slope of AB} = \frac{\text{distance}}{\text{time}} = \frac{100 - 50}{20 - 10} = \frac{50}{10} = 5$$

$$\& \quad \text{Slope of CD} = \frac{\text{distance}}{\text{time}} = \frac{300 - 250}{40 - 35} = \frac{50}{5} = 10$$

∴ Speed of CD > Speed at AB
[Slope of CD > Slope of AB]

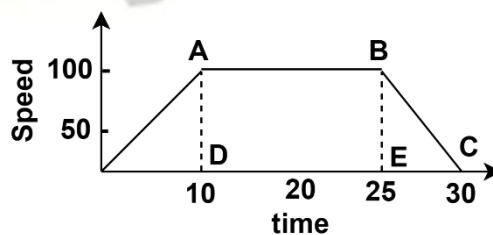
So,

This is non – uniform motion.

Q. Find the distance in the given graph.

Here,

O to A body accelerates
A to B body uniform speed
B to C body retards



$$\begin{aligned}\text{Ans Distance} &= \text{Area of Speed – time graph} \\ &= \text{area of trapezium} \\ &= \frac{1}{2} \times (\text{Sum of parallel side}) \times \text{height}\end{aligned}$$

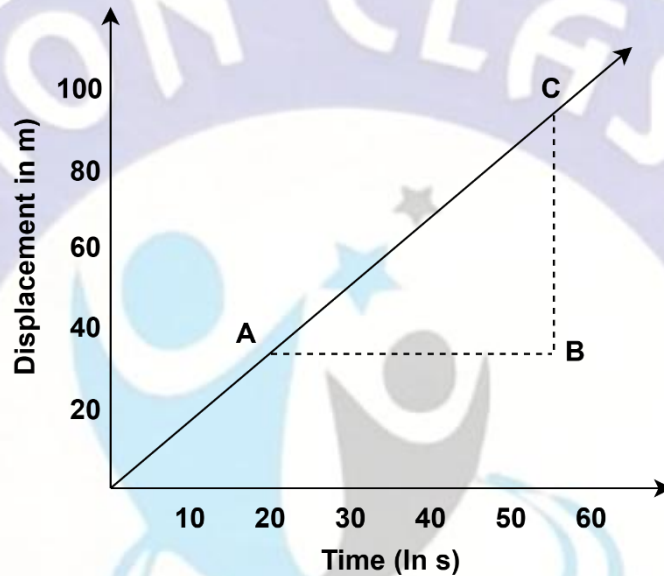
$$\begin{aligned}
 &= \frac{1}{2} \times (AB + OC) \times AD \\
 &= \frac{1}{2} \times [(25 - 10) + (30 - 0)] \times [100 - 0] \\
 &= \frac{1}{2} \times [15 + 30] \times 100 \\
 &= 45 \times 50 = 225 \text{ meter}
 \end{aligned}$$

Thus,

The body travels 225 meters.

▪ Case 5:

- The displacement – time graph of an object moving with uniform velocity is a straight line.



$$\begin{aligned}
 \text{Velocity} = \text{Slope of AC} &= \frac{BC}{AB} \\
 &= \frac{100 - 40}{60 - 20} = \frac{60}{40} \\
 &= 1.5 \text{ m/s}
 \end{aligned}$$

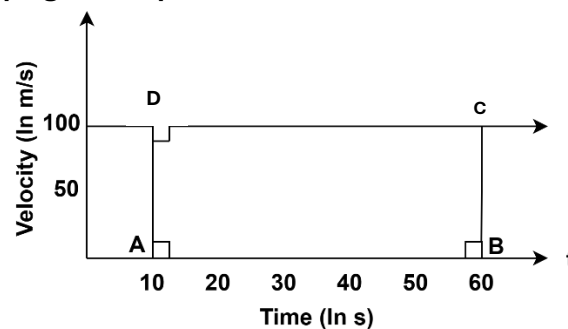
Note:

$$\text{Velocity} \propto \text{Slope} \propto \frac{y}{x}$$

i.e, The slope of displacement – time graph is as more high the velocity is a high.

▪ Case 6:

- The area of velocity – time graph gives displacement.



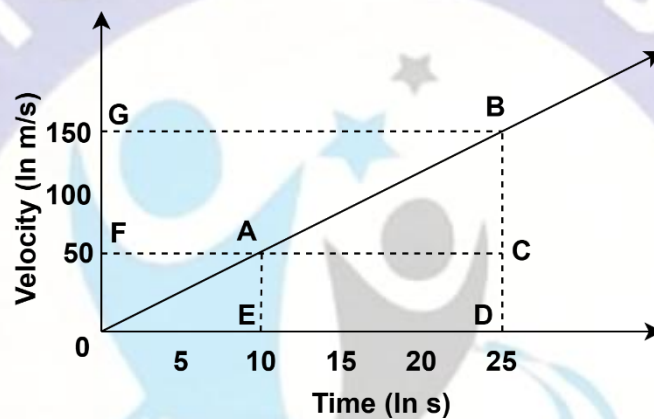
∴ Displacement = Area of rectangle ABCD
 = Length × Breadth
 = AB × AD
 = (60 – 10) × (100 – 0)
 = 50 × 100 = 5000 m

∴ Required displacement = 5000 metre

∴ other way, displacement = Velocity × time
 = (100 – 0) × (60 – 10)
 = 100 × 50 = 5000 metre

▪ Case 7:

➤ The velocity time graph gives acceleration



Acceleration = Slope of $\frac{OA}{AB}$
 = Slope of AB = $\frac{BC}{AC} = \frac{150 - 50}{25 - 10} = \frac{100}{15} = 6.67$
 = Slope of AB = $\frac{100}{15} = 6.67$

∴ Required acceleration = 6.67 m/s

Note:

Acceleration ∝ Slope of Velocity – time graph

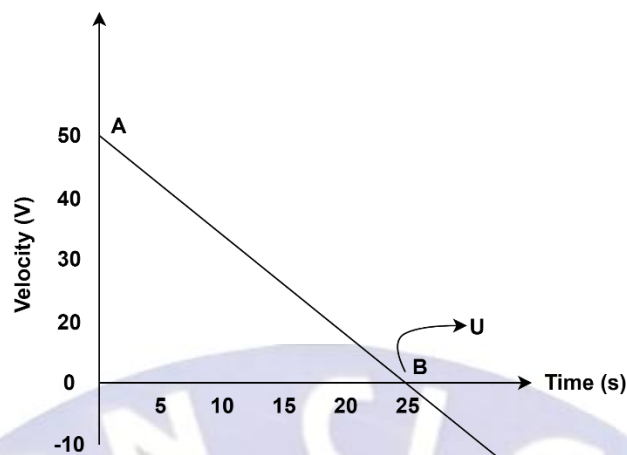
i.e.,

$\vec{a} (\uparrow)$, Slope ($\uparrow$)

$\vec{a} (\downarrow)$, Slope ($\downarrow$)

▪ Case 8:

➤ The negative Slope of Velocity – time graph gives retardation.

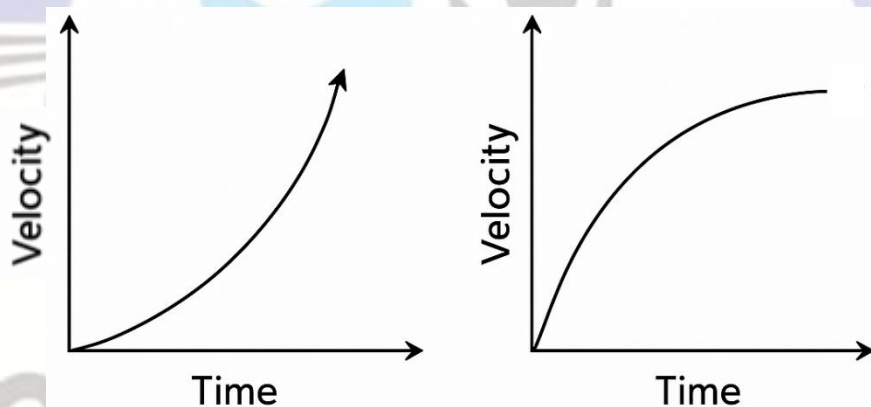


Retardation = Slope of AB = $\frac{OA}{OB} = \frac{0 - 50}{25 - 0} = \frac{-50}{25}$

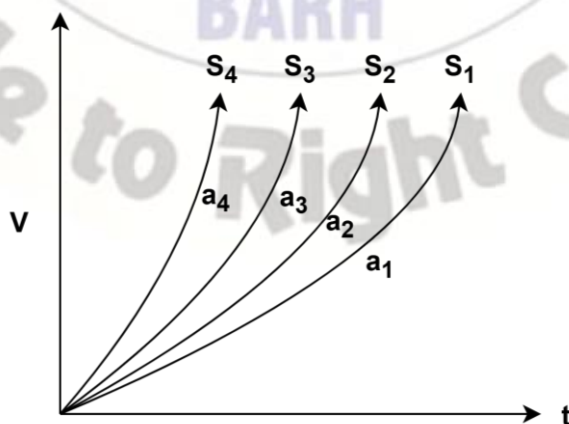
∴ Retardation = Slope of AB = -2 m/s^2

Case 9:

- The velocity – time graph of non – uniform acceleration is a curve line.



Note:



Here,

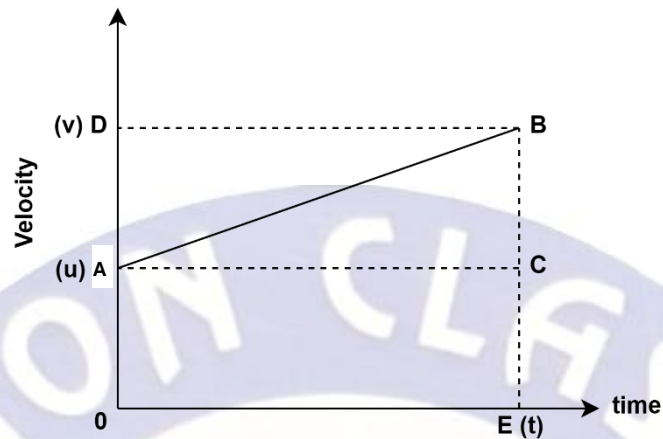
Slope of $S_4 > S_3 > S_2 > S_1$

Acceleration: $a_4 > a_3 > a_2 > a_1$

- The equation for motion in a straight line with uniform acceleration.

OR

The equation for motion.



At, $t = 0$

Initial velocity = u

$\Rightarrow OA = u$

At time t

Final velocity = v

$\Rightarrow OD = v$

- Case 1: We know that the slope of velocity – time graph gives acceleration

$$\therefore \text{Slope of AB} = \frac{BC}{AC}$$

$$\therefore a = \frac{BE - CE}{OE} = \frac{OD - OA}{OE}$$

$$\Rightarrow a = \frac{v-u}{t} \quad \Rightarrow at = v - u$$

$$\Rightarrow v = u + at$$

- Case 2: The area of velocity time graph gives displacement

$$\therefore \text{Displacement} = \text{ar (trapezium OABE)}$$

$$= \text{Area of rectangle OACE} + \text{ar. (} \triangle ABC \text{)}$$

$$= OA \times OE + \frac{1}{2} \times AC \times BC$$

$$= OA \times OE + \frac{1}{2} \times OE \times \left(\frac{BC}{AC} \right) \times AC$$

$$= OA \times OE + \frac{1}{2} \times OE \times \left(\frac{BC}{AC} \right) \times OE \quad [\because AC = OE]$$

$$S = OA \times OE + \frac{1}{2} \times \left(\frac{BC}{AC} \right) \times OE^2$$

Here,

$$OA = u, OE = t \text{ and } \frac{BC}{AC} = \text{Slope} = a$$

$$\therefore S = u \times t + \frac{1}{2} \times a \times t^2$$

OR

$$S = ut + \frac{1}{2} at^2$$

- Case 3: The area of velocity – time graph gives displacement

∴ Displacement = area of trapezium OABE

$$\therefore S = \frac{1}{2} \times (\text{Sum of parallel sides}) \times \text{height}$$

$$\begin{aligned} \Rightarrow S &= \frac{1}{2} \times (OA + BE) \times AC \\ &= \frac{1}{2} \times (OA + OD) \times AC \\ &= \frac{1}{2} \times (OA + OD) \times \left(\frac{AC}{BC} \times BC \right) \end{aligned}$$

$$\Rightarrow S = \frac{1}{2} \times (OA + OD) \times \left(\frac{BC}{AC} \right)$$

$$\Rightarrow S = \frac{1}{2} \times (OA + OD) \times \left(\frac{BE - CE}{\frac{BC}{AC}} \right)$$

$$\Rightarrow S = \frac{1}{2} \times (OA + OD) \times \left(\frac{OD - OA}{\frac{BC}{AC}} \right)$$

$$\Rightarrow S = \frac{1}{2} \times \left[\frac{OD^2 - OA^2}{\frac{BC}{AC}} \right]$$

$$\Rightarrow S = \frac{1}{2} \times \left[\frac{v^2 - u^2}{a} \right]$$

$$\Rightarrow 2as = v^2 - u^2$$

$$\Rightarrow v^2 = u^2 + 2as$$

2nd method

$$\therefore S = ut + \frac{1}{2}at^2$$

$$\Rightarrow S = u \times \left(\frac{v-u}{a} \right) + \frac{1}{2}a \times \left(\frac{v-u}{a} \right)^2$$

$$\Rightarrow S = \frac{uv - u^2}{a} + \frac{1}{2}a \times \frac{(v-u)^2}{a^2}$$

$$\Rightarrow S = \frac{uv - u^2}{a} + \frac{(v-u)^2}{2a}$$

$$\Rightarrow S = \frac{2(uv - u^2) + (v-u)^2}{2a}$$

$$\Rightarrow S = \frac{2uv - 2u^2 + v^2 + u^2 - 2uv}{2a} = \frac{v^2 - u^2}{2a}$$

$$\Rightarrow 2as = v^2 - u^2 \Rightarrow v^2 = u^2 + 2as$$

$$\therefore v = u + at$$

$$at = v - u$$

$$t = \frac{v-u}{a}$$

- Average Velocity of Uniform accelerated motion

$$\therefore V_{\text{avg}} = \frac{\text{Total displacement}}{\text{Total time}}$$

$$\therefore V_{\text{avg}} = \frac{S}{t} = \frac{1}{t} \times S$$

$$\Rightarrow V_{\text{avg}} = \frac{1}{t} \times \left(ut + \frac{1}{2}at^2 \right)$$

$$\Rightarrow V_{\text{avg}} = \frac{1}{t} \times t \left[u + \frac{1}{2}at \right]$$

$$\Rightarrow V_{\text{avg}} = u + \frac{1}{2}at$$

$$\begin{aligned} &= u + \frac{v-u}{2} \\ &= \frac{2u + v - u}{2} = \frac{u+v}{2} \end{aligned}$$

$$\begin{aligned} [\because v &= u + at \\ v - u &= at] \end{aligned}$$

$$V_{\text{avg}} = \frac{u + v}{2}$$

Special Note:

➤ If the particle moves with rest then initial velocity (u) = 0.

i) $\therefore v = u + at$
 $\Rightarrow v = 0 + at$
 $\Rightarrow v = at$

ii) $\therefore S = ut + \frac{1}{2}at^2$
 $\therefore S = 0 \times t + \frac{1}{2}at^2$
 $\Rightarrow S = \frac{1}{2}at^2$

iii) $\therefore v^2 = u^2 + 2as$
 $\Rightarrow v^2 = 0^2 + 2as$
 $\Rightarrow v^2 = 2as$

• **Uniform Circular Motion**

- When an object moves with a uniform speed on a circular path then the motion of that object is said to be uniform circular motion.
- Uniform Circular motion is an example of accelerated motion.
- The direction of an object is variable at every point of circular motion either motion is uniform or non – uniform.
- The direction of velocity in circular motion is in the direction of tangent on every point.

If Circular radius = r

Uniform speed of particle = V

The distance travelled in a round by the particle = $2\pi r$

$\therefore \text{time} = \frac{\text{distance}}{\text{speed}}$
 $\therefore t = \frac{2\pi r}{v}$

$$v = \frac{2\pi r}{t}$$